An Erdős–Kac law for palindromes and for reversed primes

Abstract

For every integer $b \geq 2$ we determine the distribution of the number of prime factors, counted with or without multiplicity, of the base-b palindromes with λ digits and of the base-b reversals of the primes with λ digits: both obey an Erdős–Kac law with the classical centring $\log\log b^\lambda$ and scaling $\sqrt{\log\log b^\lambda}$. For the number of distinct prime factors we obtain in addition all moments of order up to $\frac{1}{2}(\log\log b^\lambda)^{1/3}$, uniformly in the order. For reversed primes the law persists when the leading digit of the prime is prescribed. The arithmetic inputs are a theorem of Col for palindromes and the Bombieri–Vinogradov theorem of Dartyge, Rivat and Swaenepoel for reversed primes; in both cases the excluded primes are those dividing $b(b^2-1)$. The deduction rests on the sieve-theoretic moment estimate of Granville and Soundararajan, cast as a general criterion in the manner of Murty, Murty and Pujahari and modified to allow a finite set of primes at which the local densities are arbitrary.

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1 Introduction

Let $\omega(n)$ be the number of distinct prime factors of n and $\Omega(n)$ the number of prime factors of n counted with multiplicity. The theorem of Erdős and Kac [10] states that, uniformly for $t \in \mathbb{R}$,

$$\frac{1}{x}\,\#\Big\{n \leq x:\ \omega(n) \leq \log\log x + t\sqrt{\log\log x}\Big\} \ \longrightarrow\ \Phi(t) \qquad (x \to \infty), \tag{1.1}$$

where Φ is the standard normal distribution function. For sequences defined by conditions on the digits, laws of this kind are known in several cases. Mauduit and Sárközy proved an Erdős–Kac theorem for the integers whose digit sum lies in a fixed residue class [18], and for those whose digit sum is fixed [19, Thm. 4]. Erdős, Mauduit and Sárközy [11, Thm. 1] obtained an Erdős–Kac law for ω on the integers whose base-b digits are confined to a set $\mathcal{D} = \{0, d_2, \ldots, d_s\}$ with $\gcd(d_2, \ldots, d_s) = 1$ and $s > \sqrt{b}$. Konyagin, Mauduit and Sárközy [16] studied the normal order and the large values of ω on missing-digit sets, together with the large values of Ω on the integers with a fixed digit sum. For palindromes and for reversals of primes we are not aware of such a law.

Fix an integer $b \geq 2$. Every integer $n \geq 0$ has a unique base-b expansion $n = \sum_{j\geq 0} \varepsilon_j(n) b^j$ with digits $\varepsilon_j(n) \in \{0, 1, \ldots, b-1\}$, and we write

$$R_\lambda(n) \ :=\ \sum_{j=0}^{\lambda-1} \varepsilon_j(n)\, b^{\lambda-1-j} \tag{1.2}$$

for the reversal of the λ least significant digits of n. Let

$$\mathcal{T}_\lambda \ :=\ \{n:\ b^{\lambda-1} \leq n < b^\lambda,\ \ n = R_\lambda(n)\}$$

be the set of base-b palindromes with exactly λ digits, so that $\#\mathcal{T}_\lambda = (b-1)b^{\lceil \lambda/2\rceil -1}$; and let

$$\mathcal{P}_\lambda := \{p \text{ prime}: \ b^{\lambda-1} \le p < b^\lambda\}, \qquad \pi_\lambda := \#\mathcal{P}_\lambda,$$

so that the integers $R_\lambda(p)$ with $p \in \mathcal{P}_\lambda$ are the *reversed primes*. The term refers to the reversals themselves, which are in general composite; it is not to be confused with the *reversible primes* of [7], the primes whose reversal is again prime. This note determines the distribution of ω and of Ω on both families, the palindromes and the reversed primes.

For palindromes the question is old. Banks and Shparlinski [1], in their work on prime divisors of palindromes, showed that for every large λ some $n \in \mathcal{T}_\lambda$ has at least $(\log\log n)^{1+o(1)}$ distinct prime factors. Their result concerns the largest value of ω on $\mathcal{T}_\lambda$, and it does not address the typical value. We show in Theorem 1.1 below that almost all palindromes, and not merely one, have $(1+o(1))\log\log n$ distinct prime factors, the fluctuation around $\log\log n$ being of order $\sqrt{\log\log n}$. Concerning the typical order of ω on $\mathcal{T}_\lambda$ nothing seems to be known, although the factorisation of palindromes has been studied from other directions: Col [6] and Tuxanidy and Panario [24] produce palindromes with boundedly many prime factors, Johnston and Kerr [15] show that infinitely many palindromes are squarefree in every base, with an asymptotic count, and Chourasiya and Johnston [5] obtain cubefree palindromes in every base, and squarefree reversals of primes for $b \ge 26000$.

Reversed primes have become accessible only recently. Bhowmik and Suzuki [2, 3] established an effective Siegel–Walfisz theorem for the reversals $R_\lambda(p)$ in arithmetic progressions, first for $b \ge 31699$ and then for every $b \ge 2$; Harm and Johnston [13] have since given a version in which the number of digits is not fixed. Dartyge, Rivat and Swaenepoel [8] obtained a Bombieri–Vinogradov theorem for reversals whose moduli range up to a fixed power of b^λ; it is quoted below as Theorem 4.4. Their own application is to a sieve problem: they produce an explicit κ_b such that $\Omega(R_\lambda(p)) \le \kappa_b$ for $\gg b^\lambda\lambda^{-2}$ primes $p \in \mathcal{P}_\lambda$, with $\kappa_{10} = 1378$. It is not known whether $R_\lambda(p)$ is prime for infinitely many primes p, nor whether there are infinitely many palindromic primes.

1.1 Notation

The letters p and ℓ always denote primes, $b \ge 2$ is fixed, and $\lambda \ge 2$ is an integer parameter, which is understood throughout to tend to infinity. We set $\omega(1) = \Omega(1) = 0$, write $v_\ell(n)$ for the exponent of ℓ in the factorisation of n, and put

$$\Phi(t) \ := \ \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t} e^{-u^2/2}\, du.$$

Implied constants in the Landau and Vinogradov symbols $O(\cdot)$, $o(\cdot)$, $\ll$ and $\asymp$ may depend on b and on the level-of-distribution exponent ξ of Proposition 1.5, and in Section 2 also on δ, $\#\mathcal{E}$ and the implied constant in (1.7) (all introduced in §1.2 below); they depend on other parameters only where these are displayed as subscripts. In particular no implied constant depends on k.

1.2 Statement of results

Throughout we abbreviate

$$L := \log\log b^\lambda, \qquad C_k := \frac{\Gamma(k+1)}{2^{k/2}\,\Gamma(k/2+1)}, \tag{1.3}$$

so that C_k is the k-th moment of the standard normal law when k is even. Every n occurring in Theorems 1.1–1.3 lies in $[b^{\lambda-1}, b^\lambda)$ for $\lambda \ge 3$, as the proof of Corollary 1.4 records, so that $L = \log\log n + O(\lambda^{-1})$; the reader may therefore read L as $\log\log n$ throughout.

Theorem 1.1. *Let $b \geq 2$ be an integer. Then, uniformly for $t \in \mathbb{R}$,*

$$\frac{1}{\#\mathcal{T}_\lambda}\#\Big\{n \in \mathcal{T}_\lambda :\ \omega(n) \leq L + t\sqrt{L}\Big\} \ = \ \Phi(t) + o(1) \qquad (\lambda \to \infty),$$

and the same asymptotic holds with ω replaced by Ω.

For $1 \leq i \leq b-1$ put

$$\mathcal{P}_{\lambda,i} := \{p \in \mathcal{P}_\lambda :\ \varepsilon_{\lambda-1}(p) = i\} = \{p \text{ prime} :\ ib^{\lambda-1} \leq p < (i+1)b^{\lambda-1}\}, \qquad \pi_{\lambda,i} := \#\mathcal{P}_{\lambda,i},$$

so that $\mathcal{P}_\lambda$ is the disjoint union of the $\mathcal{P}_{\lambda,i}$, and write

$$\mathcal{A}_\lambda := \{R_\lambda(p) :\ p \in \mathcal{P}_\lambda\}, \qquad \mathcal{A}_{\lambda,i} := \{R_\lambda(p) :\ p \in \mathcal{P}_{\lambda,i}\}. \tag{1.4}$$

By (1.2), i is the last digit of $R_\lambda(p)$ for $p \in \mathcal{P}_{\lambda,i}$. Reversal of a block of λ digits is an involution, so that R_λ is injective on $\mathcal{P}_\lambda$ (see the start of Section 4); hence $\#\mathcal{A}_\lambda = \pi_\lambda$ and $\#\mathcal{A}_{\lambda,i} = \pi_{\lambda,i}$, and averages over $\mathcal{A}_\lambda$ coincide with averages over $\mathcal{P}_\lambda$.

Theorem 1.2. *Let $b \geq 2$ be an integer and let f denote either ω or Ω. Then, uniformly for $t \in \mathbb{R}$,*

$$\frac{1}{\pi_\lambda}\#\Big\{p \in \mathcal{P}_\lambda :\ f\big(R_\lambda(p)\big) \leq L + t\sqrt{L}\Big\} \ = \ \Phi(t) + o(1) \qquad (\lambda \to \infty),$$

and the same asymptotic holds for each fixed $i \in \{1, \ldots, b-1\}$ with $\mathcal{P}_\lambda$ and π_λ replaced throughout by $\mathcal{P}_{\lambda,i}$ and $\pi_{\lambda,i}$.

The localised form of Theorem 1.2 is the Erdős–Kac law for the reversals of the primes whose leading digit is prescribed. The unlocalised form follows by summing over i.

The statements for ω in Theorems 1.1 and 1.2 both follow from a moment estimate that is uniform in the order of the moment.

Theorem 1.3. *Let $b \geq 2$ and let $\mathcal{B}_\lambda$ be $\mathcal{T}_\lambda$, or $\mathcal{A}_\lambda$, or $\mathcal{A}_{\lambda,i}$ for a fixed $i \in \{1, \ldots, b-1\}$. Then, uniformly for positive integers $k \leq \frac{1}{2}L^{1/3}$, one has*

$$\frac{1}{\#\mathcal{B}_\lambda}\sum_{n \in \mathcal{B}_\lambda}\big(\omega(n) - L\big)^k \ = \ C_k\, L^{k/2}\Big(1 + O\big(k^{3/2}L^{-1/2}\big)\Big) \quad (k \text{ even}),$$

$$\frac{1}{\#\mathcal{B}_\lambda}\sum_{n \in \mathcal{B}_\lambda}\big(\omega(n) - L\big)^k \ \ll \ C_k L^{k/2} k^{3/2} L^{-1/2} \qquad (k \text{ odd}).$$

For $\mathcal{B}_\lambda$ the set of all integers below b^λ, Theorem 1.3 is contained in Theorem 1 of Granville and Soundararajan [12], from which the form of the error term is taken. The exponent $1/3$ is the one imposed by their moment estimate (Proposition 2.1 below); the explicit factor $\frac{1}{2}$ is a convenience, absorbing the difference between L and the variance of the sieve model constructed in Section 2. Theorems 1.1 and 1.2 use only the case of fixed k; the uniformity is recorded with a view to Remark 5.4. Theorem 1.3 is stated for ω only: the corresponding statements for Ω in Theorems 1.1 and 1.2 come from hypothesis (iii) of Proposition 1.5 below and are not quantitative.

Corollary 1.4. *For every $\theta > 0$ the number of $n \in \mathcal{T}_\lambda$ with $|\omega(n) - L| > \theta L$ is $o(\#\mathcal{T}_\lambda)$, and the number of $p \in \mathcal{P}_\lambda$ with $|\omega(R_\lambda(p)) - L| > \theta L$ is $o(\pi_\lambda)$; likewise for Ω. In particular, ω and Ω have normal order* $\log\log n$ *on both families.*

In particular, for each fixed K, all but $o(\pi_\lambda)$ primes $p \in \mathcal{P}_\lambda$ have $\omega(R_\lambda(p)) > K$; so the set of $p \in \mathcal{P}_\lambda$ with $\Omega(R_\lambda(p)) \le \kappa_b$ has density zero in $\mathcal{P}_\lambda$. This is consistent with the lower bound $\gg b^\lambda \lambda^{-2}$ for the cardinality of that set obtained in [8], since $\pi_\lambda \asymp b^\lambda \lambda^{-1}$. The constant κ_b invites comparison with the normal order, though the comparison is one of sizes only, Corollary 1.4 being asymptotic and carrying no information for a bounded range of λ. (For $b = 10$ one has $L > \kappa_{10} = 1378$ only when $\lambda > e^{1378}/\log 10$.) What makes the count of [8] a statement about reversals with atypically few prime factors is that it holds for every large λ with κ_b held fixed.

Theorems 1.1, 1.2 and 1.3 are all deduced from the following criterion. Let $(\mathcal{B}_\lambda)_{\lambda \ge 2}$ be non-empty finite sets of positive integers, put $N_\lambda := \#\mathcal{B}_\lambda$, and for $d \ge 1$ set

$$r_d(\lambda) \ := \ \#\{n \in \mathcal{B}_\lambda :\ d \mid n\} - \frac{N_\lambda}{d}. \tag{1.5}$$

For a finite set of primes $\mathcal{E}$ let

$$\mathcal{D}(\mathcal{E}) \ := \ \{d \ge 1 :\ \ell \mid d \ \Rightarrow \ \ell \notin \mathcal{E}\}. \tag{1.6}$$

Finally, call a set S of primes *regular of density* δ, where $0 < \delta \le 1$, if

$$\sum_{\substack{\ell \le y \\ \ell \in S}} \frac{1}{\ell} \ = \ \delta \log\log y + O(1) \qquad (y \ge 3), \tag{1.7}$$

and put

$$\omega_S(n) := \#\{\ell \in S :\ \ell \mid n\}, \qquad \Omega_S(n) := \sum_{\ell \in S} v_\ell(n),$$

so that $\omega_S = \omega$ and $\Omega_S = \Omega$ when S is the set of all primes, which is regular of density 1 by Mertens' theorem.

Proposition 1.5 (cf. [12, Prop. 3], [21, Thm. 3])**.** *Let $(\mathcal{B}_\lambda)_{\lambda \ge 2}$, N_λ, $r_d(\lambda)$ and $\mathcal{D}(\mathcal{E})$ be as above. Suppose that there exist a constant $\xi > 0$ and a finite set of primes $\mathcal{E}$, both independent of λ, such that*

(i) *$n \le b^\lambda$ for every $n \in \mathcal{B}_\lambda$, and*

(ii) *for every $A > 0$,*

$$\sum_{\substack{d \le b^{\xi\lambda},\ d \text{ squarefree} \\ d \in \mathcal{D}(\mathcal{E})}} \left| r_d(\lambda) \right| \ \ll_A \ N_\lambda\, \lambda^{-A}.$$

Let S be regular of density δ. Then, uniformly for positive integers $k \le \frac{1}{2}(\delta L)^{1/3}$, one has

$$\begin{aligned} \frac{1}{N_\lambda} \sum_{n \in \mathcal{B}_\lambda} \left(\omega_S(n) - \delta L\right)^k \ &= \ C_k (\delta L)^{k/2} \Big(1 + O\big(k^{3/2} (\delta L)^{-1/2}\big)\Big) \quad (k \text{ even}), \\ \frac{1}{N_\lambda} \sum_{n \in \mathcal{B}_\lambda} \left(\omega_S(n) - \delta L\right)^k \ &\ll \ C_k (\delta L)^{k/2} k^{3/2} (\delta L)^{-1/2} \qquad\qquad (k \text{ odd}). \end{aligned} \tag{1.8}$$

Consequently, uniformly for $t \in \mathbb{R}$,

$$\frac{1}{N_\lambda} \#\Big\{n \in \mathcal{B}_\lambda :\ \omega_S(n) \le \delta L + t\sqrt{\delta L}\Big\} \ = \ \Phi(t) + o(1) \qquad (\lambda \to \infty). \tag{1.9}$$

If in addition

(iii) $\sum_{n\in\mathcal{B}_\lambda} (\Omega(n) - \omega(n)) \ll N_\lambda,$

then (1.9) *also holds with* ω_S *replaced by* Ω_S.

The constant ξ is any exponent of distribution available for the family: for palindromes any $\xi < \beta(b)$ of Theorem 3.1, for reversed primes any $\xi < \xi_0(b)$ of Theorem 4.4. The conclusion does not depend on its value, only on its existence (Remark 5.1). Theorems 1.1–1.3 are the case in which S is the set of all primes and $\delta = 1$; Corollary 1.6 below is the general case.

Proposition 1.5 differs from [12, Prop. 3] and [21, Thm. 3] in exactly one respect: the presence of the finite set $\mathcal{E}$, at whose primes no hypothesis is imposed. That single change is what makes the two families of this note accessible. In a typical application of the moment estimate of [12] (quoted as Proposition 2.1 below) one chooses the multiplicative function h appearing there so as to model the local densities of the family; the hypothesis $0 \le h(d) \le d$ then encodes what is known about them. In the proof of Proposition 1.5 nothing is modelled. We take $h \equiv 1$, so that r_d means exactly (1.5), and hypothesis (ii) alone carries the assertion that the remainders are small on average. Both families treated here need this latitude: by Lemma 4.2(i) and Example 4.3 the local densities of $R_\lambda(p)$ at the primes dividing b are not $1/\ell$, and by Lemma 4.1(i) at those dividing b^2-1 they vanish, while for palindromes the constraints are those described after (3.2); any criterion that insists on modelling them correctly is unusable here. Since $\mathcal{E}$ is finite and independent of λ, discarding its primes shifts ω_S by $O(1)$, which is $o(\sqrt{L})$ and therefore invisible on the scale of (1.9): whatever happens at those primes cannot affect the limit law.

The proof of Proposition 1.5 runs as follows. Given k, one truncates at $y = b^{\xi\lambda/(k+1)}$, so that products of $k+1$ primes below y stay within the level $b^{\xi\lambda}$ of hypothesis (ii); the truncation is cheap because $\log\log y = L + \log\xi - \log(k+1)$, so that it moves the mean by $O(\log(2k))$ only. The primes of $\mathcal{E}$ and those exceeding y are then discarded, which by hypothesis (i) alters $\omega_S(n)$ by at most $\#\mathcal{E} + \lambda\log b/\log y \ll k = o(\sqrt{L})$. What is left is a sum over primes at which hypothesis (ii) is available, and to it one applies the estimate of Granville and Soundararajan.

Taking for S the primes with a prescribed Frobenius class gives the following.

Corollary 1.6. *Let K be a finite Galois extension of $\mathbb{Q}$, let $\mathcal{C}$ be a conjugacy class in* $\mathrm{Gal}(K/\mathbb{Q})$, *and let $S_\mathcal{C}$ be the set of primes unramified in K whose Frobenius class is $\mathcal{C}$; put $\delta := \#\mathcal{C}/[K:\mathbb{Q}]$. Then Theorems 1.1, 1.2 and 1.3 hold with ω replaced by $\omega_{S_\mathcal{C}}$ and Ω by $\Omega_{S_\mathcal{C}}$, and with L and $\sqrt{L}$ replaced by δL and $\sqrt{\delta L}$; in particular the moments are uniform for positive integers $k \le \frac{1}{2}(\delta L)^{1/3}$, with error term $O\big(k^{3/2}(\delta L)^{-1/2}\big)$.*

Proposition 1.5 is proved in Section 2. Section 3 treats palindromes and Section 4 reversed primes; the latter uses the set $\mathcal{E}_b$ of (3.2) and one identity from the proof of Lemma 3.3, but is otherwise independent of the former. Section 5 collects four remarks: on the arithmetic inputs and why a Siegel–Walfisz range would not suffice (Remark 5.1), on the extremal problem of [1] (Remark 5.2), on other levels of distribution for palindromes (Remark 5.3), and on rates of convergence (Remark 5.4).

2 An Erdős–Kac criterion

We use the sieve-theoretic moment estimate of Granville and Soundararajan in the following form. Let $\mathcal{M}$ be a finite multiset of N positive integers (we shall use it only when $\mathcal{M}$ is a set), put $N(d) := \#\{n \in \mathcal{M} : d \mid n\}$, let h be a non-negative multiplicative function with

$0 \le h(d) \le d$ for squarefree d, and write $N(d) = h(d)N/d + r_d$ for squarefree d. For a set of primes $\mathcal{R}$ put

$$\mu_{\mathcal{R}} := \sum_{\ell \in \mathcal{R}} \frac{h(\ell)}{\ell}, \qquad \sigma_{\mathcal{R}}^2 := \sum_{\ell \in \mathcal{R}} \frac{h(\ell)}{\ell}\Big(1 - \frac{h(\ell)}{\ell}\Big),$$

these being the mean and the variance of the sieve model introduced after (2.1) below; put $\omega_{\mathcal{R}}(n) := \#\{\ell \in \mathcal{R} :\ \ell \mid n\}$, as before; and, for an integer $j \ge 0$, let $\Pi_j(\mathcal{R})$ be the set of squarefree integers which are products of at most j primes of $\mathcal{R}$. The letter μ is used for this mean throughout, the Möbius function not occurring in this note.

Proposition 2.1 (Granville and Soundararajan [12, Prop. 3])**.** *Uniformly for all positive integers $k \le \sigma_{\mathcal{R}}^{2/3}$, one has*

$$\sum_{n \in \mathcal{M}} \big(\omega_{\mathcal{R}}(n) - \mu_{\mathcal{R}}\big)^k = C_k N \sigma_{\mathcal{R}}^k \Big(1 + O\Big(\frac{k^3}{\sigma_{\mathcal{R}}^2}\Big)\Big) + O\Big(\mu_{\mathcal{R}}^k \sum_{d \in \Pi_k(\mathcal{R})} |r_d|\Big) \quad (k \text{ even}),$$

$$\sum_{n \in \mathcal{M}} \big(\omega_{\mathcal{R}}(n) - \mu_{\mathcal{R}}\big)^k \ll C_k N \sigma_{\mathcal{R}}^k \frac{k^{3/2}}{\sigma_{\mathcal{R}}} + \mu_{\mathcal{R}}^k \sum_{d \in \Pi_k(\mathcal{R})} |r_d| \qquad (k \text{ odd}).$$

Throughout this section $(\mathcal{B}_\lambda)$, N_λ, ξ, $\mathcal{E}$ and S are as in Proposition 1.5, with S regular of density δ. Given a positive integer $k \le \frac{1}{2}(\delta L)^{1/3}$, set

$$y \ := \ b^{\xi\lambda/(k+1)}, \qquad \mathcal{Q} \ := \ \{\ell \le y :\ \ell \in S,\ \ell \notin \mathcal{E}\}, \qquad \mu := \mu_{\mathcal{Q}}, \quad \sigma := \sigma_{\mathcal{Q}}, \tag{2.1}$$

the last two formed with $h \equiv 1$, so that $\mu = \sum_{\ell \in \mathcal{Q}} \ell^{-1}$ and $\sigma^2 = \sum_{\ell \in \mathcal{Q}} \ell^{-1}(1 - \ell^{-1})$. These are the mean and the variance of $\sum_{\ell \in \mathcal{Q}} X_\ell$, where the X_ℓ are independent Bernoulli variables with $\mathbb{P}(X_\ell = 1) = 1/\ell$: this is the sieve model for $\omega_{\mathcal{Q}}$ in the sense of [12], and Proposition 2.1 says that the moments of $\omega_{\mathcal{Q}} - \mu$ follow it.

The next lemma records the three properties of (2.1) that the proof needs: that the truncation moves the mean by $O(\log(2k))$ only; that discarding $\mathcal{E}$ and the primes above y changes ω_S by $O(k)$; and that every modulus arising in Proposition 2.1 is covered by hypothesis (ii).

Lemma 2.2. *With the notation* (2.1) *one has, uniformly for $1 \le k \le \frac{1}{2}(\delta L)^{1/3}$:*

(i) $\mu = \delta L + O(\log(2k))$ *and* $\sigma^2 = \delta L + O(\log(2k))$*;*

(ii) $|\omega_S(n) - \omega_{\mathcal{Q}}(n)| \le (k+1)/\xi + \#\mathcal{E} \ \ll \ k$ *for every* $n \in \mathcal{B}_\lambda$*;*

(iii) $\displaystyle\sum_{d \in \Pi_{k+1}(\mathcal{Q})} |r_d(\lambda)| \ \ll_A \ N_\lambda \lambda^{-A}$ *for every* $A > 0$.

Proof. (i) By (1.7), $\sum_{\ell \le y,\ \ell \in S} \ell^{-1} = \delta \log\log y + O(1)$, and deleting the at most $\#\mathcal{E}$ primes of $\mathcal{E}$ alters this by $O(1)$; hence $\mu = \delta \log\log y + O(1)$. From $\log y = (\xi/(k+1))\lambda \log b$ we get $\log\log y = L + \log \xi - \log(k+1)$, which gives the first assertion. The second follows from $\sigma^2 = \mu - \sum_{\ell \in \mathcal{Q}} \ell^{-2} = \mu + O(1)$.

(ii) The difference is at most $\#\{\ell \in \mathcal{E} : \ell \mid n\} + \#\{\ell > y : \ell \mid n\}$. The first term is at most $\#\mathcal{E}$; for the second, $n \le b^\lambda$ by hypothesis (i) and every prime counted exceeds $y = b^{\xi\lambda/(k+1)}$, so their number is less than $\lambda \log b / \log y = (k+1)/\xi$.

(iii) Recall that $\Pi_{k+1}(\mathcal{Q})$ is the set of squarefree integers which are products of at most $k+1$ primes of $\mathcal{Q}$, and let $d = \ell_1 \cdots \ell_j \in \Pi_{k+1}(\mathcal{Q})$, the ℓ_i being distinct elements of $\mathcal{Q}$ and $j \le k+1$. Then d is squarefree; every prime factor of d lies in $\mathcal{Q}$ and hence outside $\mathcal{E}$, so that $d \in \mathcal{D}(\mathcal{E})$ by (1.6); and each $\ell_i \le y$, so that

$$d \ \le \ y^j \ \le \ y^{k+1} \ = \ b^{\xi\lambda}$$

by the choice of y in (2.1). Hence

$$\Pi_{k+1}(\mathcal{Q}) \;\subseteq\; \{d \le b^{\xi\lambda} :\ d \text{ squarefree},\ d \in \mathcal{D}(\mathcal{E})\},$$

and since the terms $|r_d(\lambda)|$ are non-negative, the sum in (iii) is at most the sum in hypothesis (ii) of Proposition 1.5, which is $\ll_A N_\lambda \lambda^{-A}$. This is what dictates the exponent in (2.1): the truncation point y is chosen so that y^{k+1} is exactly the level $b^{\xi\lambda}$ supplied by hypothesis (ii), with $k+1$ rather than k because Proposition 2.1 is applied below at every order $m \le k+1$, and $\Pi_m(\mathcal{Q}) \subseteq \Pi_{k+1}(\mathcal{Q})$ for such m. □

We can now say where the range $k \le \frac{1}{2}(\delta L)^{1/3}$ of Proposition 1.5 comes from. It arises twice. Firstly, Proposition 2.1 is available only for $k \le \sigma^{2/3}$, and by Lemma 2.2(i) the variance of the sieve model satisfies $\sigma^2 = \delta L + O(\log(2k))$. Secondly, the binomial estimate in the proof below requires $k^{3/2} \ll \sigma$, which is the same condition. Both therefore read $k \ll (\delta L)^{1/3}$, the factor $\frac{1}{2}$ absorbing the $O(\log(2k))$ discrepancy between σ^2 and δL, together with the passage from k to $k+1$ in the proof.

Proof of Proposition 1.5. Fix $k \le \frac{1}{2}(\delta L)^{1/3}$ and let $y, \mathcal{Q}, \mu, \sigma$ be as in (2.1). Apply Proposition 2.1 with $\mathcal{M} = \mathcal{B}_\lambda$, $N = N_\lambda$, $h \equiv 1$ and $\mathcal{R} = \mathcal{Q}$; the remainders r_d are then those of (1.5), and only $d \in \mathcal{D}(\mathcal{E})$ occur in the conclusion, every element of $\Pi_{k+1}(\mathcal{Q})$ being a product of primes of $\mathcal{Q}$ and hence coprime to $\mathcal{E}$. By Lemma 2.2(i) we have $\sigma^{2/3} = (\delta L)^{1/3}(1+o(1)) \ge k+1$ for all large λ, uniformly for k in the stated range, so Proposition 2.1 is available for every order $m \le k+1$.

Write $W(n) := \omega_{\mathcal{Q}}(n) - \mu$. By Lemma 2.2(i) there is a constant C, not depending on k, with $\mu \le \delta L + C\log(2k) \le L(1 + C\log(2k)/L)$, since $\delta \le 1$. Raising this to the m-th power and using $1+u \le e^u$ gives $\mu^m \le L^m \exp(Cm\log(2k)/L)$; and for $m \le k+1$ the exponent is $o(1)$, because $2k \le (\delta L)^{1/3}$ gives $\log(2k) \le \frac{1}{3}\log L$ while $m \ll L^{1/3}$, so that $m\log(2k) \ll L^{1/3}\log L = o(L)$. Thus $\mu^m \le L^m e^{o(1)}$. In particular no constant is raised here to a power that grows with k. With Lemma 2.2(iii) at $A=1$, the second error term of Proposition 2.1 is therefore $\ll L^m\lambda^{-1}$ after division by N_λ. Since $\lambda = e^L/\log b$ and $m \le k+1 \ll L^{1/3}$, this is $e^{-L(1+o(1))}$, hence smaller than any fixed power of L^{-1}. As $C_m\sigma^m \ge 1$ for λ large, and since $\sigma^2 \asymp L$, it is in particular smaller than both $C_m\sigma^m m^3\sigma^{-2}$ and $C_m\sigma^m m^{3/2}\sigma^{-1}$, and so may be absorbed into the error terms below. Hence, for $1 \le m \le k+1$,

$$\begin{aligned} \frac{1}{N_\lambda}\sum_n W(n)^m &= C_m\sigma^m\Big(1 + O(m^3\sigma^{-2})\Big) \quad && (m \text{ even}),\\ \frac{1}{N_\lambda}\sum_n W(n)^m &\ll C_m\sigma^m m^{3/2}\sigma^{-1} && (m \text{ odd}). \end{aligned} \tag{2.2}$$

In particular, for every $0 \le m \le k$,

$$\frac{1}{N_\lambda}\sum_n \big|W(n)\big|^m \;\ll\; C_m\sigma^m. \tag{2.3}$$

For $m = 0$ both sides equal 1; for $1 \le m \le k$, Lyapunov's inequality and the even case of (2.2) give

$$\frac{1}{N_\lambda}\sum_n \big|W(n)\big|^m \;\le\; \Big(\frac{1}{N_\lambda}\sum_n W(n)^{2\lceil m/2\rceil}\Big)^{m/(2\lceil m/2\rceil)} \;\ll\; C_m\sigma^m,$$

where for odd m the last step uses $C_{m+1}^{m/(m+1)} \ll C_m$, which follows from $C_m = \Gamma(m+1)/(2^{m/2}\Gamma(m/2+1))$ and Stirling's formula.

Now put $\Delta(n) := (\omega_S(n) - \omega_{\mathcal{Q}}(n)) + (\mu - \delta L)$, so that $\omega_S(n) - \delta L = W(n) + \Delta(n)$ and, by Lemma 2.2(i) and (ii), $\|\Delta\|_\infty \ll k + \log(2k) \ll k$. By the binomial theorem,

$$\frac{1}{N_\lambda}\sum_n (\omega_S(n) - \delta L)^k = \frac{1}{N_\lambda}\sum_n W(n)^k + \sum_{j=1}^{k}\binom{k}{j}\frac{1}{N_\lambda}\sum_n W(n)^{k-j}\Delta(n)^j,$$

and by (2.3) the j-th summand is $\ll \binom{k}{j}(c_1 k)^j C_{k-j}\sigma^{k-j}$, where c_1 depends only on the parameters held fixed in this section and, in particular, not on k. Stirling's formula gives $C_m \asymp (m/e)^{m/2}$ for $m \geq 1$, and $C_0 = 1$; hence $C_{k-j} \ll C_k (e/k)^{j/2}$ for $0 \leq j \leq k$, because $((k-j)/k)^{(k-j)/2} \leq 1$. So, using $\binom{k}{j} \leq k^j/j!$ and writing $c_2 := c_1\sqrt{e}$ and absorbing the implied constant just introduced,

$$\sum_{j=1}^{k}\binom{k}{j}(c_1 k)^j C_{k-j}\sigma^{k-j} \;\ll\; C_k\sigma^k \sum_{j\geq 1}\frac{1}{j!}\Big(\frac{c_2 k^{3/2}}{\sigma}\Big)^j \;\ll\; C_k\sigma^k\,\frac{k^{3/2}}{\sigma},$$

the last step because $u := c_2 k^{3/2}/\sigma = O(1)$ in the stated range, whence $e^u - 1 \ll u$. Finally, write $\sigma^2 = \delta L(1+\vartheta)$, where $\vartheta \ll \log(2k)/(\delta L)$ by Lemma 2.2(i). Then $k|\vartheta| \ll (\delta L)^{-2/3}\log(\delta L) = o(1)$ uniformly in the stated range, so

$$\sigma^k = (\delta L)^{k/2}\exp\big(\tfrac{k}{2}\log(1+\vartheta)\big) = (\delta L)^{k/2}\big(1 + O(k\log(2k)/(\delta L))\big),$$

and $k\log(2k)/(\delta L) \ll k^{3/2}(\delta L)^{-1/2}$. Combining this with (2.2) for $m = k$, and noting $k^3\sigma^{-2} \ll k^{3/2}\sigma^{-1}$ in the stated range, gives (1.8) and its odd counterpart.

For (1.9), fix k and let $\lambda \to \infty$ in (1.8): the k-th moment of $(\omega_S - \delta L)/\sqrt{\delta L}$ tends to C_k for k even and to 0 for k odd, which are the moments of the standard normal law. That law is determined by its moments, so (1.9) follows by the method of moments in the form of [4, Thm. 30.2], uniformly in t because Φ is continuous.

Assume now hypothesis (iii), and let F_λ^ω and F_λ^Ω denote the distribution functions in (1.9) formed with ω_S and with Ω_S respectively. Since $\Omega_S(n) - \omega_S(n) = \sum_{\ell\in S}\max(v_\ell(n) - 1, 0) \leq \Omega(n) - \omega(n)$, hypothesis (iii) gives $N_\lambda^{-1}\sum_n(\Omega_S(n) - \omega_S(n)) \ll 1$. Fix $\tau > 0$ and let $\eta_\tau(\lambda)$ be the proportion of $n \in \mathcal{B}_\lambda$ with $\Omega_S(n) - \omega_S(n) > \tau\sqrt{\delta L}$; by Markov's inequality applied to the non-negative function $\Omega_S - \omega_S$ we have $\eta_\tau(\lambda) \ll 1/(\tau\sqrt{\delta L})$, a bound depending on λ and τ but not on t. Now $\Omega_S \geq \omega_S$ pointwise, so for every real c the condition $\Omega_S(n) \leq c$ forces $\omega_S(n) \leq c$; and conversely $\omega_S(n) \leq c - \tau\sqrt{\delta L}$ forces $\Omega_S(n) \leq c$ unless n is one of the $\eta_\tau(\lambda)N_\lambda$ exceptions. Taking $c = \delta L + t\sqrt{\delta L}$ therefore gives

$$F_\lambda^\omega(t-\tau) - \eta_\tau(\lambda) \;\leq\; F_\lambda^\Omega(t) \;\leq\; F_\lambda^\omega(t) \qquad (t \in \mathbb{R}),$$

whence

$$\sup_t\big|F_\lambda^\Omega(t) - \Phi(t)\big| \;\leq\; \sup_t\big|F_\lambda^\omega(t) - \Phi(t)\big| + \sup_t\big|\Phi(t) - \Phi(t-\tau)\big| + \eta_\tau(\lambda),$$

in which the middle term is at most $\tau/\sqrt{2\pi}$, since $\Phi' \leq (2\pi)^{-1/2}$. Letting $\lambda \to \infty$ with τ fixed, and then $\tau \to 0$, gives (1.9) for Ω_S, uniformly in t. □

3 Palindromes

Write $\mathcal{T} := \bigcup_{\nu\geq 1}\mathcal{T}_\nu$ for the set of all base-b palindromes and, for $x \geq 1$, $a \in \mathbb{Z}$ and $q \geq 1$, put $\mathcal{T}(x) := \{n \in \mathcal{T} : n < x\}$ and $\mathcal{T}(x,a,q) := \{n \in \mathcal{T}(x) : n \equiv a \pmod q\}$. Since $\#\mathcal{T}_\nu = (b-1)b^{\lceil \nu/2\rceil - 1}$, summation over $\nu \leq \lambda$ gives

$$\#\mathcal{T}(b^\lambda) \;\asymp\; b^{\lceil\lambda/2\rceil} \;\asymp\; \#\mathcal{T}_\lambda, \tag{3.1}$$

since the family grows geometrically, so that the cumulative and the per-length counts are interchangeable; this is what makes the passage from $\mathcal{T}(x)$ to $\mathcal{T}_\lambda$ in Lemma 3.3 harmless. We use two results from the literature: a level of distribution, and an upper bound of Brun–Titchmarsh type valid for every modulus.

Theorem 3.1 (Col [6, Thm. 2])**.** *For every integer $b \geq 2$ there is a constant $\beta = \beta(b) > 0$ such that for all $A > 0$ and $\eta > 0$,*

$$\sum_{\substack{q < x^{\beta-\eta} \\ \gcd(q, b^3-b)=1}} \sup_{z \leq x} \max_{a \in \mathbb{Z}} \left| \#\mathcal{T}(z,a,q) - \frac{\#\mathcal{T}(z)}{q} \right| \ll_{b,A,\eta} \frac{\#\mathcal{T}(x)}{\log^A x}.$$

Admissible values of $\beta(b)$ are tabulated in [6, §1]: one may take $\beta(2) = 1/30$ and $\beta(10) = 1/186$, and $\beta(b) \sim 1/(6\pi b)$ as $b \to \infty$.

Theorem 3.2 (Banks and Shparlinski [1, Thm. 7])**.** *For every integer $b \geq 2$, every $\nu \geq 1$ and every integer $d \geq 1$, $\#\{n \in \mathcal{T}_\nu :\ d \mid n\} \ll_b \#\mathcal{T}_\nu\, d^{-1/2}$.*

Theorem 3.2 is stated in [1] for a fixed number of digits, which is the form used below. The two results just quoted are complementary: Theorem 3.1 is an equidistribution statement, giving the main term $\#\mathcal{T}(z)/q$ with a saving of an arbitrary power of $\log x$; but it applies only to q below the level $x^{\beta-\eta}$ and coprime to $b^3 - b$, and it supplies hypothesis (ii) of Proposition 1.5 directly. Theorem 3.2 supplies no main term and loses a square root ($d^{-1/2}$ where equidistribution would give d^{-1}), but holds for every d, with neither a coprimality condition nor a restriction of range. It is used only in the verification of hypothesis (iii), and there only for the moduli beyond the reach of Theorem 3.1. For these an upper bound suffices, since the quantity being estimated need only be shown to be $O(1)$; that is what makes the loss of a square root affordable. The exponent $-1/2$ is critical: the case $m = 2$ in the proof of Lemma 3.3 produces $\sum_\ell \ell^{-1}$, which is bounded only because the range is truncated at $b^{\xi\lambda/2} < \ell < b^{\lambda/2}$, the two cut-offs being comparable powers of b^λ; any weaker exponent would diverge there.

Since $b^3 - b = b(b^2-1)$, the coprimality condition in Theorem 3.1 is exactly $q \in \mathcal{D}(\mathcal{E}_b)$ for the set

$$\mathcal{E}_b \ := \ \{\ell :\ \ell \mid b(b^2-1)\}, \qquad \#\mathcal{E}_b = \omega\big(b(b^2-1)\big) \ll 1. \tag{3.2}$$

The set $\mathcal{E}_b$ is larger than the structure of $\mathcal{T}_\lambda$ alone would require. If λ is even then $(b+1) \mid n$ for every $n \in \mathcal{T}_\lambda$: as $\lambda - 1$ is odd, the indices j and $\lambda-1-j$ are distinct and of opposite parity, so pairing them in $n \equiv \sum_j \varepsilon_j(n)(-1)^j \pmod{b+1}$ gives $n \equiv 0$. Moreover the last digit of a palindrome equals its leading digit and is therefore non-zero, which biases divisibility by the primes dividing b. Theorem 3.1 excludes the divisors of $b-1$ as well. Proposition 1.5 imposes no hypothesis at the primes of $\mathcal{E}$, so these constraints do not enter what follows.

Lemma 3.3. *Set $\mathcal{B}_\lambda := \mathcal{T}_\lambda$ and $\mathcal{E} := \mathcal{E}_b$, and let $0 < \xi < \beta(b)$. Then hypotheses* (ii) *and* (iii) *of Proposition 1.5 hold.*

Proof. For (ii), fix ξ' with $\xi < \xi' < \beta(b)$, let $x := b^\lambda$ and let $q \leq x^\xi$ satisfy $\gcd(q, b^3-b) = 1$. Then

$$\#\{n \in \mathcal{T}_\lambda :\ q \mid n\} = \#\mathcal{T}(b^\lambda, 0, q) - \#\mathcal{T}(b^{\lambda-1}, 0, q), \qquad \#\mathcal{T}_\lambda = \#\mathcal{T}(b^\lambda) - \#\mathcal{T}(b^{\lambda-1}),$$

both endpoints being at most x. Subtracting, the main terms combine to give exactly $\#\mathcal{T}_\lambda/q$, so by (1.5)

$$|r_q(\lambda)| \ \leq \ 2 \sup_{z \leq x} \max_{a \in \mathbb{Z}} \left| \#\mathcal{T}(z,a,q) - \frac{\#\mathcal{T}(z)}{q} \right|.$$

As $q \le x^{\xi} < x^{\xi'}$, choosing $\eta := \beta(b) - \xi' > 0$ in Theorem 3.1 and summing over q gives $\sum_q |r_q(\lambda)| \ll_{b,A} \#\mathcal{T}(x)(\log x)^{-A}$, which is $\ll_{b,A} \#\mathcal{T}_\lambda \lambda^{-A}$ by (3.1). This is (ii), and in fact for all such q and not only the squarefree ones; we record the stronger form because it will be applied below to the prime-power moduli ℓ^m in the verification of hypothesis (iii).

For (iii), use $\Omega(n) - \omega(n) = \sum_{m\ge 2} \#\{\ell : \ell^m \mid n\}$, valid since a prime ℓ is counted $\max(v_\ell(n) - 1, 0)$ times on the right; it therefore suffices to show that

$$\sum_{m\ge 2} \sum_{\ell} \frac{1}{\#\mathcal{T}_\lambda} \#\{n \in \mathcal{T}_\lambda : \ \ell^m \mid n\} \ \ll\ 1.$$

Every prime power occurring divides some $n < b^\lambda$, so $\ell < b^{\lambda/m}$ and $2 \le m \le M := \lambda \log b/\log 2$. If $\ell^m \le b^{\xi\lambda}$ and $\ell \nmid b^3 - b$, then the bound just proved for (ii) applies to the modulus ℓ^m, and after division by $\#\mathcal{T}_\lambda$ the total contribution is at most $\sum_{m\ge 2}\sum_\ell \ell^{-m} + O(\lambda^{-1}) = O(1)$. In every remaining case we appeal to Theorem 3.2, which gives $\#\{n \in \mathcal{T}_\lambda : \ell^m \mid n\} \ll \#\mathcal{T}_\lambda\, \ell^{-m/2}$. The primes $\ell \mid b^3 - b$ then contribute $\ll \sum_{\ell \mid b^3-b} \sum_{m\ge 2} \ell^{-m/2} \ll 1$, there being $O(1)$ of them; and for the pairs with $\ell^m > b^{\xi\lambda}$ we have $\ell > b^{\xi\lambda/m}$, whence

$$\begin{aligned}\sum_{2\le m\le M}\ \sum_{b^{\xi\lambda/m}<\ell<b^{\lambda/m}} \ell^{-m/2} \ &\ll\ \sum_{b^{\xi\lambda/2}<\ell<b^{\lambda/2}} \frac{1}{\ell} \ +\ \sum_{3\le m\le M} \big(b^{\xi\lambda/m}\big)^{1-m/2} \\ &\ll\ \log\frac{1}{\xi} + \lambda\, b^{-\xi\lambda/6} \ \ll\ 1.\end{aligned}$$

Here the first sum is $\log\log b^{\lambda/2} - \log\log b^{\xi\lambda/2} + o(1) = \log(1/\xi) + o(1)$ by Mertens' theorem, the upper cut-off $\ell < b^{\lambda/2}$ keeping it bounded; in the second sum $\big(b^{\xi\lambda/m}\big)^{1-m/2} = b^{-\xi\lambda(m-2)/(2m)} \le b^{-\xi\lambda/6}$ because $(m-2)/(2m) \ge \frac{1}{6}$ for $m \ge 3$, while the number of terms is $M \ll \lambda$. Adding the three contributions gives (iii). ☐

Proof of Theorem 1.1, and of Theorem 1.3 for palindromes. We apply Proposition 1.5, taking $\mathcal{B}_\lambda := \mathcal{T}_\lambda$, $\mathcal{E} := \mathcal{E}_b$ and $0 < \xi < \beta(b)$, with S the set of all primes, so that $\omega_S = \omega$ and $\delta = 1$. Hypothesis (i) holds because every element of $\mathcal{T}_\lambda$ is less than b^λ, while hypotheses (ii) and (iii) are Lemma 3.3. ☐

4 Reversed primes

We verify the hypotheses of Proposition 1.5 for the sets $\mathcal{A}_{\lambda,i}$ of (1.4). Reversing a block of λ base-b digits, leading zeros retained, is an involution on the set of such blocks; hence R_λ is injective on $\{0, \dots, b^\lambda - 1\}$, and in particular on $\mathcal{P}_\lambda$, so that $\#\mathcal{A}_{\lambda,i} = \pi_{\lambda,i}$ for $\lambda \ge 2$.

4.1 Local structure of the reversal

Reversal is not multiplicative, but it is compatible with reduction modulo $b^2 - 1$ and modulo the divisors of b. The congruences of Lemma 4.1 are those of (3.8) in [8], and its gcd assertion corresponds to (3.9) there. Stating the first congruence modulo b^2-1 rather than modulo $b-1$ and $b+1$ separately is a genuine strengthening when b is odd, since $\operatorname{lcm}(b-1, b+1) = (b^2-1)/2$ then; it also yields the equality of greatest common divisors.

Lemma 4.1 (cf. [8, (3.8), (3.9)])**.** *Let $\lambda \ge 2$ and let n be an integer with $b^{\lambda-1} \le n < b^\lambda$.*

(i) *One has $R_\lambda(n) \equiv b^{\lambda-1} n \pmod{b^2-1}$; in particular $R_\lambda(n) \equiv n \pmod{b-1}$ and $R_\lambda(n) \equiv (-1)^{\lambda-1} n \pmod{b+1}$. Moreover* $\gcd(R_\lambda(n), b^2-1) = \gcd(n, b^2-1)$, *and consequently* $\gcd(R_\lambda(p), b^2-1) = 1$ *for every $p \in \mathcal{P}_\lambda$ with $p > b+1$. One also has $b^{\lambda-1} \le R_\lambda(p) < b^\lambda$ for every $p \in \mathcal{P}_\lambda$ with $p > b$.*

(ii) *If $\ell \mid b$ then $R_\lambda(n) \equiv \varepsilon_{\lambda-1}(n) \pmod{\ell}$.*

Proof. (i) Modulo b^2-1 we have $b^2 \equiv 1$, hence $b^{-1} \equiv b$ and $b^{-j} \equiv b^j$ for every $j \ge 0$; and $\gcd(b, b^2-1) = 1$, so the powers of b are invertible. Thus (1.2) gives

$$R_\lambda(n) \;=\; b^{\lambda-1}\sum_{j=0}^{\lambda-1}\varepsilon_j(n)b^{-j} \;\equiv\; b^{\lambda-1}\sum_{j=0}^{\lambda-1}\varepsilon_j(n)b^{j} \;=\; b^{\lambda-1}n \pmod{b^2-1}.$$

Reducing modulo $b-1$ and modulo $b+1$ and using $b \equiv 1$, respectively $b \equiv -1$, gives the two stated congruences; and since $b^{\lambda-1}$ is invertible modulo b^2-1, the displayed congruence yields $\gcd(R_\lambda(n), b^2-1) = \gcd(n, b^2-1)$, which is 1 for a prime $p > b+1$. Finally $R_\lambda(p) < b^\lambda$ is immediate from (1.2), while the leading digit of $R_\lambda(p)$ is $\varepsilon_0(p) = p \bmod b$, non-zero because $\gcd(p, b) = 1$; hence $R_\lambda(p) \ge b^{\lambda-1}$.

(ii) If $\ell \mid b$ then $b^{\lambda-1-j} \equiv 0 \pmod{\ell}$ for $j < \lambda-1$, so (1.2) leaves only the term with $j = \lambda-1$. □

By (1.6) and (3.2), $d \in \mathcal{D}(\mathcal{E}_b)$ if and only if $\gcd(d, b(b^2-1)) = 1$. We next compute the local densities of $R_\lambda(p)$ at the primes dividing b; those dividing b^2-1 were settled in Lemma 4.1(i).

Lemma 4.2. *One has $\pi_{\lambda,i} \sim \pi_\lambda/(b-1)$ for each fixed $i \in \{1,\dots,b-1\}$; in particular $\pi_{\lambda,i} \asymp \pi_\lambda \asymp b^\lambda\lambda^{-1}$. Moreover, if $\ell \mid b$ then*

(i) $\frac{1}{\pi_\lambda}\#\{p \in \mathcal{P}_\lambda :\ \ell \mid R_\lambda(p)\} \;\longrightarrow\; \frac{1}{b-1}\left\lfloor \frac{b-1}{\ell}\right\rfloor \qquad (\lambda \to \infty)$,

(ii) $\displaystyle\sum_{m\ge 2}\frac{1}{\pi_\lambda}\#\{p \in \mathcal{P}_\lambda :\ \ell^m \mid R_\lambda(p)\} \;\ll\; 1.$

Proof. For fixed i the interval $[ib^{\lambda-1}, (i+1)b^{\lambda-1})$ has length comparable to its endpoints, so the prime number theorem in the form $\pi(z) \sim z/\log z$ already gives

$$\pi_{\lambda,i} = \pi\big((i+1)b^{\lambda-1}\big) - \pi\big(ib^{\lambda-1}\big) \;\sim\; \frac{b^{\lambda-1}}{(\lambda-1)\log b},$$

Here the ratio $(i+1)/i$ of the endpoints is bounded away from 1 in terms of b alone, so the difference of the two counts is comparable to each of them and the $o(1)$ relative errors are absorbed. Summing over i gives $\pi_\lambda \sim (b-1)b^{\lambda-1}/((\lambda-1)\log b)$, whence $\pi_{\lambda,i} \sim \pi_\lambda/(b-1)$: the leading digit of a prime in $\mathcal{P}_\lambda$ is asymptotically uniformly distributed on $\{1,\dots,b-1\}$.

For (i), Lemma 4.1(ii) shows that for $\ell \mid b$ one has $\ell \mid R_\lambda(p)$ if and only if ℓ divides the leading digit $\varepsilon_{\lambda-1}(p)$; exactly $\lfloor (b-1)/\ell \rfloor$ of the possible leading digits $i \in \{1,\dots,b-1\}$ qualify, so summing $\pi_{\lambda,i} \sim \pi_\lambda/(b-1)$ over these i gives (i).

For (ii), write $v := v_\ell(b) \ge 1$ and, for $m \ge 1$, put $j := \lceil m/v \rceil$, so that $\ell^m \mid b^j$. Suppose first that $j \le \lambda/2$. By (1.2), $R_\lambda(n) \equiv \sum_{r<j}\varepsilon_{\lambda-1-r}(n)b^r \pmod{\ell^m}$, so the condition $\ell^m \mid R_\lambda(n)$ depends only on the j leading digits of n; and as $(\varepsilon_{\lambda-1},\dots,\varepsilon_{\lambda-j})$ runs through $\{0,\dots,b-1\}^j$ the quantity $\sum_{r<j}\varepsilon_{\lambda-1-r}b^r$ runs bijectively through $\{0,\dots,b^j-1\}$, so the number of admissible leading blocks is at most $b^j\ell^{-m}+1$. Each fixed leading block confines p to an interval of length $b^{\lambda-j} \ge b^{\lambda/2}$, in which the Brun–Titchmarsh inequality [20, Thm. 2] gives $\ll b^{\lambda-j}/((\lambda-j)\log b) \ll \pi_\lambda b^{-j}$ primes, since $(\lambda-j)^{-1} \le 2\lambda^{-1}$ and $\pi_\lambda \asymp b^\lambda\lambda^{-1}$. Hence $\#\{p \in \mathcal{P}_\lambda : \ell^m \mid R_\lambda(p)\} \ll \pi_\lambda(\ell^{-m} + b^{-j})$, and summing over those $m \ge 2$ with $\lceil m/v \rceil \le \lambda/2$ contributes $\ll \sum_{m\ge2}\ell^{-m} + v\sum_{j\ge1}b^{-j} \ll 1$.

Suppose now that $j > \lambda/2$, so that $m > v(\lambda/2-1) \ge \lambda/2-1$. Since R_λ is injective on $\mathcal{P}_\lambda$, the values $R_\lambda(p)$ with $p \in \mathcal{P}_\lambda$ are distinct positive integers below b^λ, so that

$$\#\{p \in \mathcal{P}_\lambda :\ d \mid R_\lambda(p)\} \;\le\; \#\{1 \le n < b^\lambda :\ d \mid n\} \;\le\; b^\lambda/d \qquad (d \ge 1), \tag{4.1}$$

so, using $\pi_\lambda \asymp b^\lambda\lambda^{-1}$ and $\ell \geq 2$, the remaining m contribute $\ll \lambda\sum_{m>\lambda/2-1}\ell^{-m} \ll \lambda\, 2^{-\lambda/2+1} \ll 1$. $\square$

Example 4.3. For $b = 10$ we have $b(b^2-1) = 2\cdot 3^2\cdot 5\cdot 11$, so $\mathcal{E}_{10} = \{2,3,5,11\}$. By Lemma 4.1(i), $3 \nmid R_\lambda(p)$ and $11 \nmid R_\lambda(p)$ for every prime $p > 11$, while by Lemma 4.2(i) the densities of the events $2 \mid R_\lambda(p)$ and $5 \mid R_\lambda(p)$ tend to $4/9$ and $1/9$ rather than to $1/2$ and $1/5$.

4.2 Level of distribution and proof of the theorems

Write $\pi_\lambda(z)$ for the number of p with $b^{\lambda-1} \leq p < z$, and $\overline{\pi}_\lambda(z,a,d)$ for the number of those p with $R_\lambda(p) \equiv a \pmod d$, residues being taken in $\{1,\dots,d\}$, so that $a = d$ is the class $0 \bmod d$; thus $\pi_\lambda = \pi_\lambda(b^\lambda)$.

Theorem 4.4 (Dartyge, Rivat and Swaenepoel [8, Thm. 1.3])**.** *For every integer $b \geq 2$ there is an explicit constant $\xi_0 = \xi_0(b) > 0$ such that, given $\xi \in (0,\xi_0)$, there exist $c = c(b,\xi) > 0$ and $\lambda_0(b,\xi) \geq 2$ with the following property. For every integer $\lambda \geq \lambda_0(b,\xi)$,*

$$\sum_{\substack{d\leq b^{\xi\lambda}\\ \gcd(d,\,b(b^2-1))=1}} \ \sup_{b^{\lambda-1}\leq z\leq b^\lambda}\ \sup_{1\leq a\leq d}\left|\ \overline{\pi}_\lambda(z,a,d) - \frac{\pi_\lambda(z)}{d}\ \right| \ \ll\ b^{\lambda-c\sqrt{\lambda}}.$$

The supremum over z is what permits the restriction to $\mathcal{P}_{\lambda,i}$ in the next lemma: for $\mathcal{A}_\lambda$ itself the single value $z = b^\lambda$ would serve, so the localised form of Theorem 1.2 is bought precisely by that supremum.

Lemma 4.5. *Let $\xi \in (0,\xi_0(b))$ and let $i \in \{1,\dots,b-1\}$. Then for every $A > 0$,*

$$\sum_{\substack{d\leq b^{\xi\lambda}\\ d\in\mathcal{D}(\mathcal{E}_b)}} |r_d(\lambda)| \ \ll_A\ \pi_{\lambda,i}\,\lambda^{-A},$$

where $r_d(\lambda)$ is formed from $\mathcal{A}_{\lambda,i}$ as in (1.5)*, and the sum is over all such d and not only the squarefree ones.*

Proof. Since $\mathcal{P}_{\lambda,i}$ consists of the p with $ib^{\lambda-1} \leq p < (i+1)b^{\lambda-1}$,

$$\#\{n \in \mathcal{A}_{\lambda,i} :\ d \mid n\} = \overline{\pi}_\lambda((i+1)b^{\lambda-1},d,d) - \overline{\pi}_\lambda(ib^{\lambda-1},d,d),$$

$$\pi_{\lambda,i} = \pi_\lambda((i+1)b^{\lambda-1}) - \pi_\lambda(ib^{\lambda-1}),$$

both endpoints lying in $[b^{\lambda-1},b^\lambda]$. Subtracting, the main terms combine to give exactly $\pi_{\lambda,i}/d$, so $|r_d(\lambda)|$ is at most twice the supremum appearing in Theorem 4.4, and that theorem gives $\sum_d |r_d(\lambda)| \ll b^{\lambda-c\sqrt{\lambda}}$ with $c = c(b,\xi) > 0$. By Lemma 4.2, $\pi_{\lambda,i} \asymp b^\lambda\lambda^{-1}$, so the right-hand side is $\pi_{\lambda,i}\cdot O(\lambda b^{-c\sqrt{\lambda}})$; and $b^{-c\sqrt{\lambda}} = \exp(-c\sqrt{\lambda}\log b)$ decays faster than any fixed power of λ^{-1}. $\square$

Lemma 4.6. *For every $i \in \{1,\dots,b-1\}$ one has $\sum_{p\in\mathcal{P}_{\lambda,i}}(\Omega(R_\lambda(p)) - \omega(R_\lambda(p))) \ll \pi_{\lambda,i}$.*

Proof. Since $\mathcal{P}_{\lambda,i} \subseteq \mathcal{P}_\lambda$ and $\pi_{\lambda,i} \asymp \pi_\lambda$ by Lemma 4.2, it suffices to prove the bound with $\mathcal{P}_\lambda$ and π_λ in place of $\mathcal{P}_{\lambda,i}$ and $\pi_{\lambda,i}$; and by the identity $\Omega(n) - \omega(n) = \sum_{m\geq 2}\#\{\ell :\ \ell^m \mid n\}$ used in the proof of Lemma 3.3 it suffices to show that

$$\sum_{m\geq 2}\sum_{\ell}\frac{1}{\pi_\lambda}\#\{p\in\mathcal{P}_\lambda :\ \ell^m \mid R_\lambda(p)\} \ \ll\ 1.$$

Fix $\xi \in (0, \xi_0(b))$ and split the range of (ℓ, m) into four parts. Every prime power counted divides some $R_\lambda(p) < b^\lambda$, so $2 \le m \le M := \lambda \log b / \log 2$ throughout.

If $\ell \mid b^2 - 1$ then $\ell \nmid R_\lambda(p)$ for $p > b+1$ by Lemma 4.1(i), so these ℓ contribute $O(1)$ in total; and if $\ell \mid b$ then Lemma 4.2(ii) bounds the corresponding subsum by $O(1)$ for each of the $O(1)$ such ℓ.

Let now $\ell \nmid b(b^2-1)$. If $\ell^m \le b^{\xi\lambda}$ then $\ell^m \in \mathcal{D}(\mathcal{E}_b)$, so Lemma 4.5 applies. Distinct pairs (ℓ, m) with $m \ge 2$ give distinct moduli ℓ^m, so summing the conclusion of that lemma over the $b-1$ values of i and using $\#\{p \in \mathcal{P}_\lambda : \ell^m \mid R_\lambda(p)\} = \pi_\lambda \ell^{-m} + \sum_i r^{(i)}_{\ell^m}(\lambda)$, where $r^{(i)}$ is formed from $\mathcal{A}_{\lambda,i}$, we obtain

$$\sum_{m \ge 2} \sum_{\substack{\ell \nmid b(b^2-1) \\ \ell^m \le b^{\xi\lambda}}} \frac{1}{\pi_\lambda} \#\{p : \ \ell^m \mid R_\lambda(p)\} \ \le \ \sum_{m \ge 2} \sum_{\ell} \frac{1}{\ell^m} + O(\lambda^{-1}) \ = \ O(1).$$

Finally, still with $\ell \nmid b(b^2-1)$, if $\ell^m > b^{\xi\lambda}$ then $\ell > u_m := b^{\xi\lambda/m}$, and (4.1) with $\pi_\lambda \asymp b^\lambda \lambda^{-1}$ and $\sum_{n>u} n^{-m} \le u^{-m} + u^{1-m}/(m-1) \le 2u^{1-m}$, valid for $u \ge 1$ and $m \ge 2$, gives

$$\begin{aligned} \sum_{2 \le m \le M} \sum_{\ell > u_m} \frac{1}{\pi_\lambda} \#\{p : \ \ell^m \mid R_\lambda(p)\} \ &\ll \ \lambda \sum_{2 \le m \le M} \sum_{\ell > u_m} \frac{1}{\ell^m} \\ &\ll \ \lambda \sum_{2 \le m \le M} u_m^{1-m} \ \ll \ \lambda^2 b^{-\xi\lambda/2} \ \ll \ 1, \end{aligned}$$

since $u_m^{1-m} = b^{-\xi\lambda(m-1)/m} \le b^{-\xi\lambda/2}$ for $m \ge 2$ and the sum over m has $O(\lambda)$ terms. $\square$

Proof of Theorem 1.2, and of Theorem 1.3 for reversed primes. Fix $i \in \{1, \dots, b-1\}$ and apply Proposition 1.5 to $\mathcal{B}_\lambda := \mathcal{A}_{\lambda,i}$ with $\mathcal{E} := \mathcal{E}_b$, $\xi \in (0, \xi_0(b))$ fixed, and S the set of all primes.[1] Hypothesis (i) holds by (1.2), which gives $R_\lambda(p) < b^\lambda$ for every $p \in \mathcal{P}_\lambda$; hypothesis (ii) is Lemma 4.5, and hypothesis (iii) is Lemma 4.6. This gives Theorem 1.3 for $\mathcal{A}_{\lambda,i}$ and the localised form of Theorem 1.2.

Since $\mathcal{P}_\lambda$ is the disjoint union of the $\mathcal{P}_{\lambda,i}$, every average over $\mathcal{A}_\lambda$ is the convex combination, with weights $\pi_{\lambda,i}/\pi_\lambda$, of the corresponding averages over the $\mathcal{A}_{\lambda,i}$. Applying this to $(\omega(n) - L)^k$ gives Theorem 1.3 for $\mathcal{A}_\lambda$, the $b-1$ estimates having the same main term and the same relative error when k is even, and the same upper bound when k is odd; and applying it to the distribution functions of the localised statement, $F_{\lambda,i}(t) := \pi_{\lambda,i}^{-1} \#\{p \in \mathcal{P}_{\lambda,i} : \ \omega(R_\lambda(p)) \le L + t\sqrt{L}\}$, gives $\sup_t |F_\lambda - \Phi| \le \max_i \sup_t |F_{\lambda,i} - \Phi| = o(1)$ for the corresponding function F_λ formed with $\mathcal{P}_\lambda$ and π_λ, which is the unlocalised form of Theorem 1.2. The argument for Ω is identical. $\square$

Proof of Corollaries 1.4 and 1.6. For Corollary 1.4, apply Theorems 1.1 and 1.2 with $t = \pm\theta\sqrt{L}$, where $L \to \infty$, and use $\Phi(-u) + 1 - \Phi(u) \to 0$ as $u \to \infty$. For the last assertion, every element of $\mathcal{T}_\lambda \cup \mathcal{A}_\lambda$ lies in $[b^{\lambda-1}, b^\lambda)$ for $\lambda \ge 3$: for $\mathcal{T}_\lambda$ by definition, and for $\mathcal{A}_\lambda$ by Lemma 4.1(i), since $p \ge b^{\lambda-1} \ge b^2 > b$ for every $p \in \mathcal{P}_\lambda$. On that range $\log\log n = L + O(\lambda^{-1})$, as was already noted after (1.3). For Corollary 1.6, let $S := S_{\mathcal{C}}$ be the set of primes unramified in K whose Frobenius class is $\mathcal{C}$; by the Chebotarev density theorem with the classical error term [17, Thm. 1.3] and partial summation, $\sum_{\ell \le y, \ \ell \in S} \ell^{-1} = \delta \log\log y + O(1)$, so S is regular of density δ, and the proofs above apply verbatim with this S. $\square$

[1] By Lemma 4.2 we have $\pi_{\lambda,i} > 0$ for all large λ; we set $\mathcal{B}_\lambda := \{1\}$ for the finitely many λ at which $\mathcal{A}_{\lambda,i}$ is empty. This does not affect the limits, since all statements are asymptotic in λ.

5 Remarks

Remark 5.1. The proofs require a level of distribution with moduli up to a fixed power of b^λ, together with the treatment of the primes dividing $b(b^2-1)$ in (3.2) and Lemma 4.2. Any improvement of $\xi_0(b)$ in Theorem 4.4, or of $\beta(b)$ in Theorem 3.1, leaves the results unchanged. A Siegel–Walfisz range for the moduli would not suffice: for reversed primes the range $d \le \exp(c\sqrt{\lambda})$ of [2, 3], and for palindromes the range $q \le \exp(c\lambda/\log\lambda)$ of [6, Thm. 1], would force the truncation point y of (2.1) below every fixed power of b^λ, and the bound $\#\{\ell > y : \ell \mid n\} \le \lambda \log b/\log y$ of Lemma 2.2(ii) would then exceed $\sqrt{L}$.

Remark 5.2. Theorem 1.1 determines the typical value of ω on $\mathcal{T}_\lambda$ and says nothing about its largest value, which is the question of [1] and remains open: the trivial upper bound is $\max_{n\in\mathcal{T}_\lambda} \omega(n) \ll \lambda/\log\lambda$, while the lower bound of [1] is $(\log\log n)^{1+o(1)} = \lambda^{o(1)}$. The two are far apart, and Theorem 1.1 does not narrow the gap. It does recover the lower bound in the form in which it is stated, and for a positive proportion of $n \in \mathcal{T}_\lambda$ rather than for a single one: for each fixed t the proportion of $n \in \mathcal{T}_\lambda$ with $\omega(n) \ge \log\log n + t\sqrt{\log\log n}$ tends to $1 - \Phi(t) > 0$, and $\log\log n + t\sqrt{\log\log n} = (\log\log n)^{1+o(1)}$.

Remark 5.3. Tuxanidy and Panario [24, Thm. 1.5] prove a level of distribution $1/5 - \eta$ for palindromes, with an exponent independent of b, whereas Col's $\beta(b) \sim 1/(6\pi b)$ degrades as b grows. Their result is stated for the palindromes coprime to $b^3 - b$, and every palindrome with an even number of digits is divisible by $b + 1$; the set to which it applies therefore contains no element of $\mathcal{T}_\lambda$ when λ is even. Substituting it for Theorem 3.1 in Lemma 3.3 would give Theorem 1.1 for the palindromes in $\mathcal{T}_\lambda$ coprime to $b^3 - b$, with ξ close to $1/5$, but only for λ odd. For the squarefree palindromes an exponent of distribution $1/16$ is due to Tuxanidy [23].

Remark 5.4. Theorem 1.3 is quantitative, but we do not deduce from it a rate of convergence in Theorem 1.1 or Theorem 1.2. The classical method of moments [4, Thm. 30.2] is qualitative, and the known routes to a rate for ω proceed instead through characteristic functions, as in Rényi and Turán [22], or through Stein's method, as in Harper [14]. For a digitally defined sequence a rate has been obtained by the former route: Mauduit and Sárközy [19, Thm. 4] give $\ll (\log\log\log N)/\sqrt{\log\log N}$ for the integers up to N with a fixed digit sum. A cumulant criterion such as [9, Thm. 2.4] does convert into a Kolmogorov bound, but requires control of cumulants of every order.

Remark 5.5 (Formal verification)**.** The deductive content of this note has been verified with the Lean 4 proof assistant (Lean 4.33.0 with Mathlib). In the companion repository

`https://github.com/vibefrtz/vibemath`

(folder `erdos-kac-reversed-primes`, archived at `doi:10.5281/zenodo.22078540`)

the results quoted from the literature — Proposition 2.1, Theorems 3.1, 3.2 and 4.4, the Brun–Titchmarsh inequality, the prime number theorem, Mertens' second theorem and the method of moments — are declared as axioms, in the form in which they are used here, and everything this note itself proves is machine-checked from those axioms with no unproved goals: Proposition 1.5 together with Lemma 2.2 and the moment computation; Lemmas 3.3, 4.1, 4.2, 4.5 and 4.6; and the deductions of Theorems 1.1 and 1.2 (pointwise and uniformly in t, localised and unlocalised), of Theorem 1.3 at each fixed moment order, and of Corollaries 1.4 and 1.6, the latter for an arbitrary regular set S. An axiom audit included in the repository lists, for every theorem, the exact axioms its proof consumes; these coincide with the citations invoked by the corresponding proof in this note. Two points lie outside the verification and are documented in the repository: the transcriptions of the quoted results were not checked mechanically against their sources beyond what is recorded there, and the uniformity in k asserted in Theorem 1.3 is verified at each fixed k only — the case that Theorems 1.1 and 1.2 use.

Declaration on the use of AI tools

This note was prepared with extensive assistance from large language models (Claude Fable 5 and Claude Opus 5, Anthropic), used at every stage of its preparation: for the literature search, for jointly working out the main arguments, and for drafting and typesetting the manuscript itself. In particular, Theorem 3.1, on which Theorem 1.1 depends, was located by the models, and the modification of [12, Prop. 3] presented in Proposition 1.5 was worked out jointly with the models. The author has verified all statements, proofs and references, made the final decisions on content and presentation, and is responsible for any errors that remain.

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